

Chapter 1: Basics of Calculus & Rate of Change Notes

Class 10 Advanced Mathematics • Calculus Fundamentals & Exam Review

1. Introduction to Limits & Continuity

Calculus is the mathematical study of continuous change. This section introduces the concept of limits, which forms the foundation of derivatives.

- **The Limit Concept:** Evaluates the value that a function approaches as the input approaches some value. Formulated as: $\lim_{x \rightarrow c} f(x) = L$.
- **Continuity:** A function $f(x)$ is continuous at $x = c$ if the limit exists, $f(c)$ is defined, and the limit equals $f(c)$.

2. Derivative as Rate of Change

The derivative measures the instantaneous rate of change of a function. Geometrically, it is the slope of the tangent line to the curve.

- **First Principles definition:** $f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h$.
- **Power Rule:** $d/dx [x^n] = n * x^{(n-1)}$.
- **Sum and Difference Rule:** $d/dx [f(x) \pm g(x)] = f'(x) \pm g'(x)$.

3. Sample Exam Review Questions

Q1. [Calculus - 3 Marks]

Find the limit: $\lim_{x \rightarrow 3} (x^2 - 9) / (x - 3)$. [Solution Hint: Factor the numerator as $(x-3)(x+3)$]

Q2. [Derivatives - 5 Marks]

Differentiate the function $f(x) = 4x^3 - 5x^2 + 7x - 10$ with respect to x . Find the slope of the tangent at point $x = 2$.